Ch 4.1-4.3: Classification

Lecture 7 - CMSE 381

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Announcements

Last time:

- Lab on Linear Regression

Announcements:

- Third homework due Friday! Covers:
  - Fri 1/28 Lab
  - Mon 1/31 Lecture
  - Weds 2/2 lecture
- Office hours - ONLINE
  - This week only: No OH Thursday

Covered in this lecture

- Classification basics
- Logistic regression
- Confounding variables

Section 1

Classification Overview

What is classification

X_1
X_2
$\vdots$

Classification: When the response variable is qualitative

↳ unordered set

- Given feature vector X and qualitative response Y in the set S, the goal is to find a function (classifier) $C(X)$ taking X as input and predicting its value for Y.
- We are more interested in estimating the probabilities that X belongs to each category

Ex) eye color $\in$ {brown, blue, green, ...}
1/3 1/3 1/3

email $\in$ {spam, not spam}

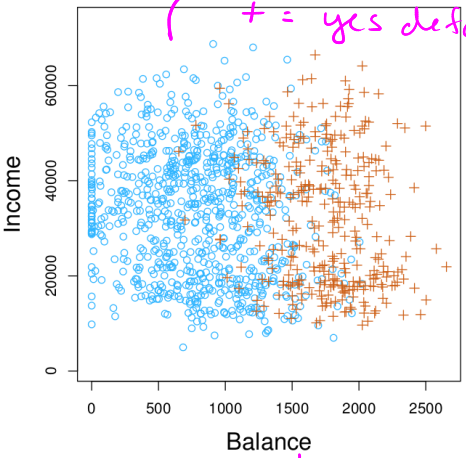
$\vdots$

Some examples

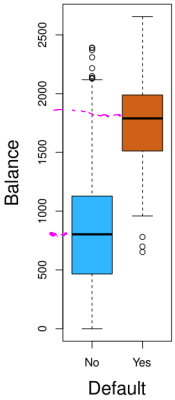
- Predict whether a COVID19 vaccine will work on a patient given patient's age $Y =$ yes/no X_1
- An online banking service wants to determine whether a transaction being performed is fraudulent on the basis of the user's IP address, past transactions, etc. $Y =$ yes/no X_1 X_2

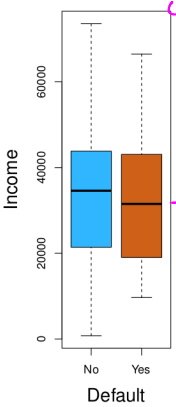
Simulated Default data set

0 = no default
+ = yes default



Not really correlated





Loan

C(Balance, Income) = yes/no

Let's just use regression!

JK that's a bad idea

Bad idea:

- Set Y to be a dummy variable taking values in $\{0, 1, 2, \cdots\}$
- Run regression, and choose k based on what integer value $\hat{y}$ is closest to

Ex.

Implies proximity

$$Y = \begin{cases} 1 & \text{if stroke} \\ 2 & \text{if drug overdose} \\ 3 & \text{if epileptic seizure} \end{cases}$$

vs.

$$Y = \begin{cases} 1 & \text{if mild} \\ 2 & \text{if moderate} \\ 3 & \text{if severe} \end{cases}$$

Bad idea is still not a great idea for two levels

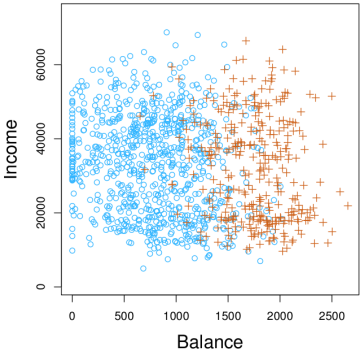
$$Y = \begin{cases} 0 & \text{if stroke} \\ 1 & \text{if overdose} \end{cases}$$

- Fit linear regression
- Predict overdose if $\hat{y} > 0.5$; stroke otherwise

- Probability? But lin reg can take values outside of [0, 1]
- Can't deal w/ more than 2 levels

Section 2

Logistic Regression



Goal: Model the probability that Y belongs to a particular category

Ex.

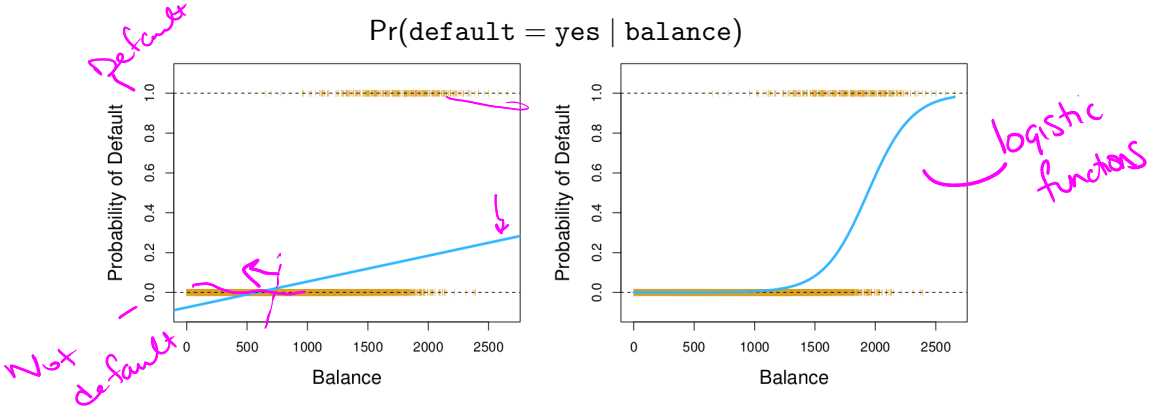
$\Pr(\texttt{default} = \texttt{yes} \mid \texttt{balance})$

$= p(\text{balance})$

Choose threshold based Comfort w/ risk

Approximating the probability

$$\Pr(\texttt{default} = \texttt{yes} \mid \texttt{balance})$$



Logistic Regression

- Let $p(X) = \Pr(Y = 1 \mid X)$.
- Turn this into something with range $\mathbb{R}$

$$\ln\left(\overbrace{\frac{p(X)}{1-p(X)}}^{\text{odds}}\right) = \overbrace{\beta_0 + \beta_1 X}$$

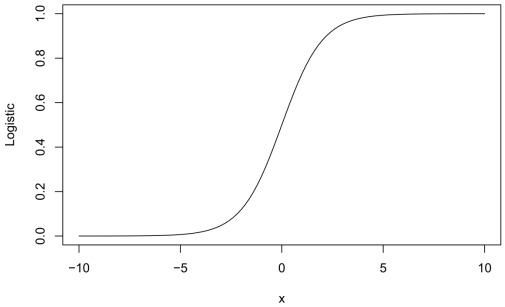
- Called *log odds*, or *logit*
- Solving for $p(X)$ gets

$$p(X) = \frac{e^{\beta_0+\beta_1 X}}{1+e^{\beta_0+\beta_1 X}}$$

logistic function

Logistic function

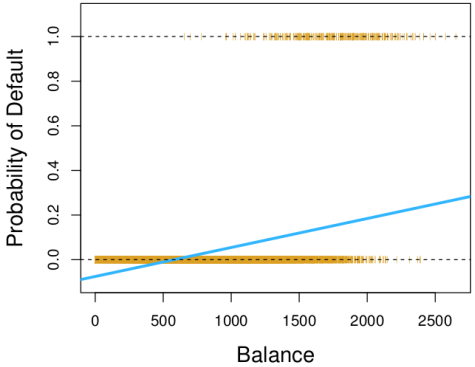
$$y = \frac{e^x}{1 + e^x}$$



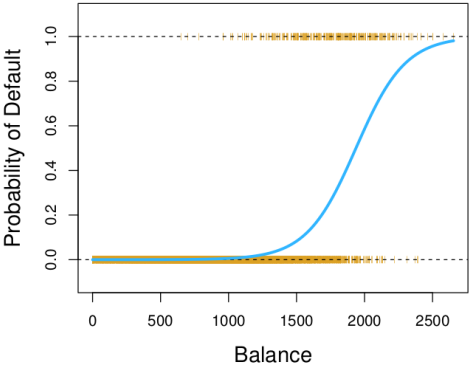
$$p(X) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}}$$

Logistic Regression

$$\Pr(\texttt{default} = \texttt{yes} \mid \texttt{balance}) = \frac{e^{\beta_0 + \beta_1 \texttt{balance}}}{1 + e^{\beta_0 + \beta_1 \texttt{balance}}}$$



Linear Regression



Logistic Regression

Estimating Coefficients: Maximum Likelihood Estimation

- **Likelihood:** Probability that data is generated from a model

fixed

$$\ell(model) = \mathbb{P}[data \mid model]$$

RV, dist.

- Find the most likely model

$$\max_{model} \ell(model)$$

Coin flip ex

If saw 50/100 heads
↳ high prob. fair coin

If saw 10/100 heads
↳ low chance coin fair

- Hard to maximize likelihood, instead maximize log

$$\max_{model} \log(\ell(model))$$

- Strictly increasing log function doesn't change maximum

Maximum Likelihood for Logistic Regression

model

$$\Pr(Y = 1 \mid X) = p(X) = \frac{e^{\beta_0+\beta_1 X}}{1 + e^{\beta_0+\beta_1 X}}$$

data pts

$$\ell(\beta_0, \beta_1) = \prod_{i|y_i=1} p(x_i) \prod_{i'|y_{i'}=0} (1 - p(x_{i'}))$$

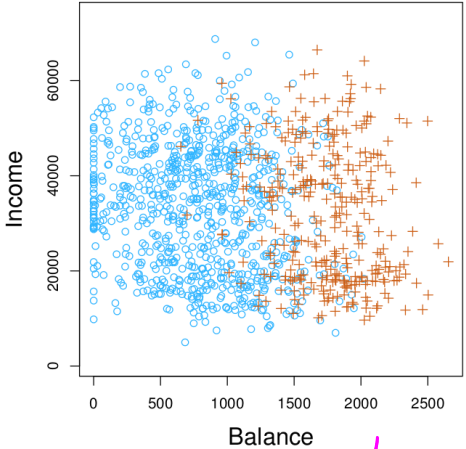
Typo? No β_i's ???

Find β_0, β_1 maximize

That's $\hat{\beta}_0, \hat{\beta}_1$

Computer magic

Default data set example



	Coefficient	Std. error	z-statistic	p-value
Intercept	−10.6513	0.3612	−29.5	<0.0001
balance	0.0055	0.0002	24.9	<0.0001

β_1

- β_1 positive: Increase in balance increases chance of default
- Z-stat: Hypothesis test
  Null: $H_0: \beta_1 = 0$
  likely relationship, b/c small pval
  $\Rightarrow \beta_1 \neq 0$
- Intercept not interesting.
  Purpose: adjusts avg fitted prob to proportion of def in (n)

Using coefficients to make predictions

	Coefficient	Std. error	z-statistic	p-value
Intercept	-10.6513	0.3612	-29.5	<0.0001
balance	0.0055	0.0002	24.9	<0.0001

Estimated probability of default for someone with a balance of \$1,000:

$$\hat{p}(X) = \frac{e^{\hat{\beta}_0+\hat{\beta}_1 X}}{1+e^{\hat{\beta}_0+\hat{\beta}_1 X}} = \frac{e^{-10.6513+0.0055\times 1000}}{1+e^{-10.6513+0.0055\times 1000}} = 0.006$$

0.6% Chance of default

Estimated probability of default for someone with a balance of \$2,000:

$$\hat{p}(X) = \frac{e^{\hat{\beta}_0+\hat{\beta}_1 X}}{1+e^{\hat{\beta}_0+\hat{\beta}_1 X}} = \frac{e^{-10.6513+0.0055\times 2000}}{1+e^{-10.6513+0.0055\times 2000}} = 0.586$$

58.6% Chance of default

Multiple Logistic Regression

Multiple features:

$$Pr(Y=1|X) = p(X) = \frac{e^{\beta_0+\beta_1 X_1+\cdots+\beta_p X_p}}{1+e^{\beta_0+\beta_1 X_1+\cdots+\beta_p X_p}}$$

Equivalent to:

$$\log\left(\frac{p(X)}{1-p(X)}\right) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$$

	Coefficient	Std. error	z-statistic	p-value
`Intercept`	−10.8690	0.4923	−22.08	<0.0001
`balance`	0.0057	0.0002	24.74	<0.0001 ← Small!
`income`	0.0030	0.0082	0.37	0.7115
`student[Yes]`	−0.6468	0.2362	−2.74	0.0062 ← Small

Input

Student status

$$\texttt{default} = \beta_0 + \beta_1 \cdot \texttt{student}$$

	Coefficient	Std. error	z-statistic	p-value
Intercept	−3.5041	0.0707	−49.55	<0.0001
student[Yes]	0.4049	0.1150	3.52	0.0004

higher chance of defaulting if student

-vs-

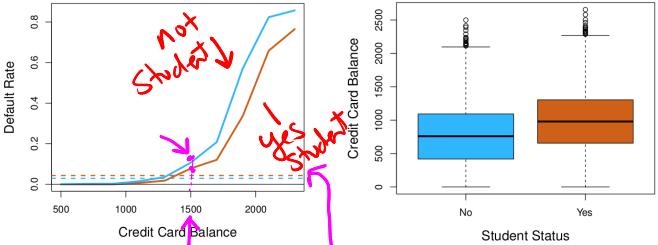
$$\texttt{default} = \beta_0 + \beta_1 \cdot \texttt{balance} + \beta_2 \cdot \texttt{income} + \beta_3 \cdot \texttt{student}$$

	Coefficient	Std. error	z-statistic	p-value
Intercept	−10.8690	0.4923	−22.08	<0.0001
balance	0.0057	0.0002	24.74	<0.0001
income	0.0030	0.0082	0.37	0.7115
student[Yes]	−0.6468	0.2362	−2.74	0.0062

lower chance of defaulting if student

Status vs Balance

Confounding variables: affect other variables in a way that produces spurious or distorted associations between two variables.



not student

yes student

· fix Balance

a student is less likely to default than a non-student

Avg'd over the whole the students have higher prob of defaulting

This causes the positive coeff in single variable version

Multinomial Logistic Regression

What if we have a categorical variable with more than two levels (let's say K)?

3

Plan A

Play the dummy variable game:

Make K the baseline:

p=K (?)

$$\Pr(Y = k|X = x) = \frac{e^{\beta_{k0}+\beta_{k1}x_1+\cdots+\beta_{kp}x_p}}{1+\sum_{l=1}^{K-1} e^{\beta_{l0}+\beta_{l1}x_1+\cdots+\beta_{lp}x_p}}$$

k=1, ..., K-1

$$\Pr(Y = K|X = x) = \frac{1}{1+\sum_{l=1}^{K-1} e^{\beta_{l0}+\beta_{l1}x_1+\cdots+\beta_{lp}x_p}}.$$

Works like before, but need to just be careful about choice of baseline and interpretation of coefficients

affects intercept info

k=1, ..., K

Plan B: Softmax coding

Treat all levels symmetrically

$$\Pr(Y = k|X = x) = \frac{e^{\beta_{k0}+\beta_{k1}x_1+\cdots+\beta_{kp}x_p}}{\sum_{l=1}^{K} e^{\beta_{l0}+\beta_{l1}x_1+\cdots+\beta_{lp}x_p}}.$$

$$\log\left(\frac{\Pr(Y = k|X = x)}{\Pr(Y = k'|X = x)}\right) = (\beta_{k0} - \beta_{k'0}) + (\beta_{k1} - \beta_{k'1})x_1 + \cdots + (\beta_{kp} - \beta_{k'p})x_p.$$

↳ Investigation in HW

Next time

- Weds:
  - 4.4: Generative models for classification
- Thurs:
  - No office hours due to conflict, send message with issue or go to other available times!
- Fri:
  - Data ethics lab
  - HW #3 due. Covers:
    - 1/28: Linear Regression Lab
    - 1/31: 4.1-4.3 Logstic Regression
    - 2/2: 4.4 Generative models

Yuck that class was a mess

↳ Very short (30 minutes under)

↳ Next time could add more about the MLE stuff to add content

↳ Last slide was BS, I couldn't figure out the eqns on the fly, could use way more examples.

First day of in person, no idea if that caused any of it